

PRAKTIKUM BASIS DATA

MODUL 09

Jurnal



**Universitas
Telkom**

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JURNAL

Running [code sql](#) berikut, screenshot hasilnya dan analisis hasil code tersebut

1. create table employees as select * from hemployees.
2. select * from employees.
- 3.

```
1 CREATE OR REPLACE PROCEDURE increase_salary (  
2 department_id_in IN employees.department_id%TYPE,  
3 increase_pct_in IN NUMBER)  
4 IS  
5 BEGIN  
6 FOR employee_rec  
7 IN (SELECT employee_id  
8 FROM employees  
9 WHERE department_id = increase_salary.department_id_in)  
10 LOOP  
11 UPDATE employees emp  
12 SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in  
13 WHERE emp.employee_id = employee_rec.employee_id;  
14 DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);  
15 END LOOP;  
16 END increase_salary;  
17
```

4. BEGIN
increase_salary (90, .50);
ROLLBACK;
END;
/
select * from employees where department_id = 90
5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.

```
-- CREATE TABLE hemployees (  
-- employee_id NUMBER(6) NOT NULL,  
-- first_name VARCHAR2(20),  
-- last_name VARCHAR2(25) NOT NULL,  
-- email VARCHAR2(25) NOT NULL,  
-- phone_number VARCHAR2(20),  
-- hire_date DATE NOT NULL,  
-- job_id VARCHAR2(10) NOT NULL,  
-- salary NUMBER(8,2),  
-- commission_pct NUMBER(2,2),  
-- manager_id NUMBER(6),  
-- department_id NUMBER(4),  
-- CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)  
-- );  
  
-- INSERT data untuk department_id = 90 (Executive -- dipakai soal no.4)  
INSERT INTO hemployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-2003','DD-MON-YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);  
INSERT INTO hemployees VALUES (101, 'Meena', 'Kochhar', 'MKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005','DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);  
INSERT INTO hemployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-2001','DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);  
  
-- INSERT tambahan department lain (agar SELECT * tidak kosong)  
INSERT INTO hemployees VALUES (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', TO_DATE('03-JAN-2006','DD-MON-YYYY'), 'IT_PROG', 9000, NULL, 102, 60);  
INSERT INTO hemployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', TO_DATE('21-MAY-2007','DD-MON-YYYY'), 'IT_PROG', 6000, NULL, 103, 60);  
INSERT INTO hemployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007','DD-MON-YYYY'), 'IT_PROG', 4200, NULL, 103, 60);  
INSERT INTO hemployees VALUES (124, 'Evin', 'Mourgos', 'EMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007','DD-MON-YYYY'), 'ST_MAN', 5800, NULL, 100, 50);  
INSERT INTO hemployees VALUES (141, 'Trenna', 'Rajs', 'TRAJS', '650.121.8009', TO_DATE('17-OCT-2003','DD-MON-YYYY'), 'ST_CLERK', 3500, NULL, 124, 50);  
INSERT INTO hemployees VALUES (149, 'Eleni', 'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29-JAN-2008','DD-MON-YYYY'), 'SA_MAN', 10500, .2, 100, 80);  
INSERT INTO hemployees VALUES (174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267', TO_DATE('11-MAY-2004','DD-MON-YYYY'), 'SA_REP', 11000, .3, 149, 80);
```

SELECT * FROM employees;

Output :

Query Result x

 All Rows Fetched: 10 in 0.008 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	80

SELECT * FROM employees WHERE department_id = 90;

Output :

Script Output x Query Result x

 All Rows Fetched: 3 in 0.011 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

Analisis :

Eksperimen ini menunjukkan bahwa manipulasi data dapat dilakukan dengan lebih efisien dan terorganisir melalui penggunaan *procedure*. Di sisi lain, kontrol transaksi melalui perintah *ROLLBACK* memegang peranan krusial karena kemampuannya untuk membatalkan seluruh modifikasi yang telah terjadi. Dengan demikian, penguasaan terhadap mekanisme kontrol transaksi menjadi faktor penentu dalam menjamin terjaganya konsistensi informasi di dalam basis data.